Last time we talked about confidence intervals

We want to use data to estimate some unknown quantity of interest, and confidence intervals give a plausible range of values for the unknown quantity, based on the observed data.

CI: plausible range

In particular, we discussed CIs for proportions & means.

Based on earlier lectures, we know sample means and sample proportions are (approximately) normally distributed

$\hat{p}$, $\bar{x}$ approx normal

Generally: collect sample data → sample mean tends to be within 2 standard deviations (SD) of true (population) mean

$\bar{x}$ within ±2 SD of $\mu$ w/prob 95% 95% of the time,

because it follows a normal distribution.

Based on that idea, we said it was plausible that the true mean $\mu$ is within ±2SD of the sample mean we observed,

idea: $\mu \in \bar{x} \pm 2^*SD$, "95% confident" which we described as being "95% confident" the true mean $\mu$ is in that interval, $\bar{x} - 2^*SD$ to $\bar{x} + 2^*SD$.

There was nothing special about the 95% level - can do other % CIs, just need to check how many SD away from the mean we need to go to capture that much of the probability of the normal distribution

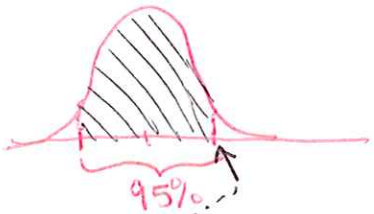
other % CI - check percentiles of N(0,1)

Additional complication - when estimating a mean, sometimes we don't know the population stdev, and instead we estimate it from the data.

don't know $\sigma^2$
↳ estimate from data
don't write $S_x^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$ $\sigma^2 \approx$ sample variance

When we estimate the stdev from the data, we use the t distrib., instead of Normal, to construct CI

AND: use t dist. instead of Normal




Under the normal dist, to get a 95% CI, we go ±2 SD out from the mean. Now, need to know how many SDs to go, under the t distribution, to get 95% area under the curve.

t param: df = n-1

The t dist. has a parameter called "degrees of freedom", which equals n-1 for this type of problem.

qt(0.975, df = n-1)

Today we are studying hypothesis tests, another tool for using observed data to learn about unknown quantity of interest.

Introduce this with an example.
In 2005, study found that 60% of drivers in MS (state) wear seat belts.

2005 study: % seat belt wearers = 60%

$p$ = % seatbelt wearers in 2025 = ?

We're interested in the % who wear seatbelts in 2025, which we don't know.

- observe $n=30$ drivers
- could: give CI for $p$

We go observe 30 drivers & check seatbelt usage.

Could construct CI to give range of plausible values for $p$.

But suppose we're specifically interested in checking whether the percentage is still 60%, the value in 2005. This type of question is the basis for a hypothesis test.

Q: Is $p=0.6$ still true?

"$p=0.6$" : hypothesis to test

We don't know $p$. We observe 30 randomly selected drivers, 23 were wearing seat belts. Based on that data, could it be true that the population percentage is 60%?

observe $n=30$, $\hat{p} = \frac{23}{30} \approx 77\%$

Is data compatible w/hypothesis? Want to know: is the observed data compatible with the true % being 60%? And, want to quantify that compatibility.

$$p=0.6 \Rightarrow \hat{p} \approx N\left(0.6, \frac{(0.6)(1-0.6)}{30}\right)$$

If $p=0.6$, then for random samples of 30 drivers, we know the sample proportion is approx Normal w/ expected value 0.6 and variance $\frac{p(1-p)}{n}$, ie standard dev. $\sqrt{\frac{0.6(0.4)}{30}} \approx 0.09$.

This lets us ask: if true $p=0.6$, is it unusual to see a sample of 30 people w/ $\hat{p} = 77\%$?

If $p=0.6$, is $\hat{p}=.77$ reasonable?

$\mathbb{P}(\geq 17\%$ away from expected val$) = ?$

Know probability dist of sample proportion → can check the prob. of getting a sample proportion at least 17 percentage points away from the expected 60%.

Compare to normal dist

$\hat{p} \approx N(0.6, 0.09^2)$

$\hat{p} = 0.77$ : how many SD away from mean?

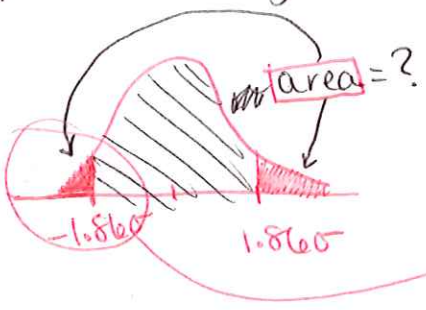
To check, we need to compare against the normal distrib.

How many SD away from the mean was our observed 77% sample proportion?

$\frac{0.77 - 0.6}{0.09}$ } distance / } size of 1 SD $\approx 1.86$

↳ $\approx 1.86$ standard deviations.

In a normal dist, what is prob of being that far away from the mean?




pnorm(-1.86, 0, 1) $\approx 0.03$

Use cumulative distrib function to get area of left tail, then multiply by 2 to get the total red shaded area (it's symmetric).

$\approx 0.03 * 2 \approx 0.06$

If truly $p=0.6$, then the observed data was 1.86 SD away from expected value. The prob. of getting a sample proportion $\geq 1.86$ SD away from the expected value is 6%

If 60% of all MS drivers wear seat belts, it would be pretty unusual to see data like the sample data we got. The data we observed isn't very compatible with the ~~idea~~ hypothesis of the true percentage being 60%.

data don't agree w/ $p = 0.6$ hypothesis., think $p \neq 0.6$

But we <u>did</u> get this sample data, so based on that, we don't think it's very plausible that the true percentage is 60%.

This is the general format of a hypothesis test

test format: state null hypothesis $H_0$ $H_0: p = 0.6$

observe data $\hat{p} = 0.77$

Then we ask, how different is the observed data from what we would expect to see if $H_0$ were true.

does data agree w/ $H_0$?

p value $\approx 0.06$

What is the probability of seeing data at least this "unusual" if $H_0$ is true? That's the p value. Quantifies how well the data "agrees" with $H_0$

~~Standard~~ decision: small p value → reject $H_0$

Based on the p value, we state our conclusion: If the p value is small, ~~data~~ the observed data doesn't agree with the null hypothesis, and our conclusion is that we "reject" $H_0$. Don't think it's true.

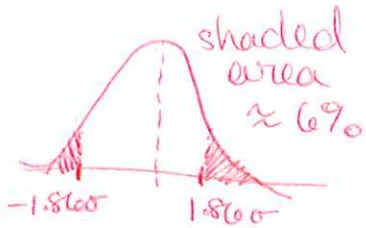
To actually calculate the p value, we checked how many SD away from the expected value our observed sample proportion was.

want: $P(|\hat{p} - 0.6| > 0.17)$ (p)

$\frac{0.77 - 0.6}{0.09} \approx 1.86$

It was 1.86 SD away, and then we checked the prob. of being at least 1.86 SD away from the mean in a normal distribution.

$P(\text{normal}(0,1) > 1.86 \text{ or } \text{normal}(0,1) < -1.86)$




That probability was approx. 6%, so our actual observed 77% sample proportion would be a pretty unlikely result in a world where the true percentage is 60%. So it probably isn't true that 60% of people wear seatbelts in 2025.

Q: when is p value "small enough" to reject?

There is no universal rule. You might use 10% as the cutoff, or 5%, or 1%. There is always some chance that you just happened to get a weird data sample. It's good practice to choose your cutoff before performing the test.

↳ up to you
↳ decide before doing test.

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Now let's look at a different example & explore some of the different choices we can make, as well as some potential complications.

Suppose I think my toaster oven is calibrated wrong, and runs @ higher temp than manufacturer says.

$X$ = oven temp @ any moment

$X \sim N(\mu, 20^2)$

Ovens have some natural temp. fluctuation, so at any point in time, the temp. is normally distributed around the set value, with SD = 20°F. (~11°C)

The manufacturer says the "bake" setting has a mean temp of 350°F, but I think it's actually hotter than that. (~175°C)

Claim: $\mu = 350$ I think: $\mu > 350$

We can frame this as a hypothesis testing problem. The null hypothesis is $\mu = 350$. The alternative I have in mind is $\mu > 350$.

$H_0: \mu = 350$ $H_A: \mu > 350$

Note: In the seatbelt example, we didn't explicitly state the alternative we had in mind. $H_0$ was $p = 0.6$, and we wanted to know if that could be true. Implicitly, the alternate was just $p \neq 0.6$

seat belt example
$H_0: p = 0.6$, $H_a: p \neq 0.6$
"two sided"

That's called a "two-sided" $H_a$, because we're open to the possibility that $p > 0.6$ or $p < 0.6$.

The oven temp is a "one-sided" $H_a$, because we're specifically asking whether $\mu > 350$.

Next we observe data. Take measurement w/thermometer @ 5 randomly chosen times.

- $X_1, \ldots, X_5 \overset{iid}{\sim} N(\mu, 20^2)$
  $\sim N(350, 20^2)$ under $H_0$

Under the null hypothesis, that $\mu = 350$, the dist. of the sample mean is

under $H_0$: $\bar{X} \sim N(350, \frac{20^2}{5})$

observe: $\bar{X} = 375$ We take the measurements, and the sample mean is 375°F. Based on that data, do we think it's really true that the oven is calibrated to run @ avg temp of 350°F?

↳ does this agree w/ $H_0$? ($\mu = 350$)

We can calculate the prob. of getting a sample avg at least 25° hotter, if the true mean is 350:

$\bar{X} \sim N(350, \frac{20^2}{5})$ 25° = how many SD?

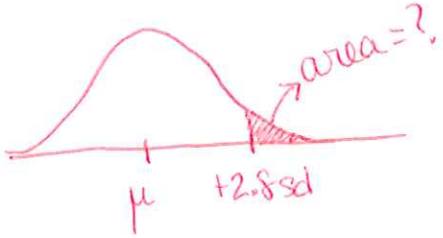
How many SD away from the expected value was our observed result?

$1\ SD = \frac{20}{\sqrt{5}} \approx 8.9$

$\frac{375 - 350}{20/\sqrt{5}} \approx 2.8$

Our sample mean was about 2.8 SD above the expected value, in a world where the true mean is 350° ($\mu$).

What is the prob of being at least 2.8SD above the mean, in a normal dist?




$P(\text{normal}(0,1) > 2.8)$

$1 - \text{pnorm}(2.8, \text{mean}=0, \text{sd}=1) \approx 0.003$

So if the oven really was calibrated to 350°, we'd be very unlikely to see a sample mean like the one we actually got. The p value is very small, the data are not compatible w/ $H_0$. small p value → data doesn't support $H_0$

In this case, we reject $H_0$, and conclude the avg oven temp is above 350°.

reject $H_0$, conclude $\mu > 350$.

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Just like w/ confidence intervals, there is a complication when we don't know the variance of the population dist.

our example: $X \sim N(\mu, 20^2)$ what if $X \sim N(\mu, \sigma^2)$, $\sigma^2$ unknown?

↳ estimate from data

↳ $\sigma^2 \approx$ sample variance $S_x^2$

What if instead, X is normally dist. w/ UNknown variance $\sigma^2$?

"how many SD away is 375 from 350?" Then when we ask how many SD away our observed $\bar{X}$ was from the expected value, we don't know how big 1 SD is. Have to estimate it from the data, using sample variance to approximate $\sigma^2$.

Suppose for our sample data, the sample variance was 500.

suppose $S_x^2 = 500$ $\bar{X}_5 \overset{\text{approx}}{\sim} N(350, \frac{500}{5})$ $sd(\bar{X}) \approx 10$

Then under the null hypothesis, the sample mean is normally dist w/ mean 350 and variance $\frac{500}{5} = 100$, so SD = 10.

We take 5 measurements, get avg of 375. How many SD away from the expected value is that?

$$\frac{375 - 350}{10} \quad \begin{matrix} \} \text{ distance} \\ \} \text{ SD} \end{matrix} \qquad = 2.5$$

So our observed data was roughly 2.5 SD away from the true mean. (population)

Then we need to answer, what is the prob of being at least 2.5 SD above the expected value?

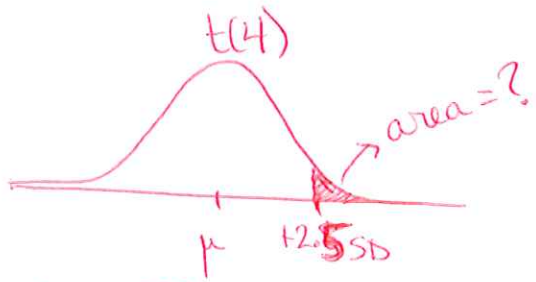
$\sigma^2$ est. $\sigma^2$ from data: use t dist, not Normal

Just like for confidence intervals, the fact that we estimated $\sigma^2$ from the data, means that we have to use the t distribution instead of normal.

$P(\geq 2.5$ SD above mean) in t dist $= ?$

In particular, the t dist with n-1 degrees of freedom, ie df = 4 for this example.

$df = n-1 = 4$




don't write $P(t(4) \geq 2.5$ SD above mean))

$$1 - pt(2.5, df=4) \approx 0.033$$

The probability is around 3%.

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One more thing about oven temp example: our probability calculations were based on sample mean being normally distributed.

we relied on $\bar{X} \sim$ Normal

We knew $\bar{X}$ was normal, because we said that the individual temp measurements X ~~were~~ themselves were normal.

we said $X \sim N(\mu, \sigma^2)$

which $\Rightarrow \bar{X} \sim$ Normal

In general, if ~~it~~ we don't know that X is normally distributed, we can still say that the SAMPLE MEAN

is approximately normal due to CLT, IF sample size is large.

X not normal

$\bar{X}_n \approx$ normal by CLT, IF n large

In this example we only had 5 data points, so if X wasn't normal, that might not have been a good approximation.

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Final point about hypothesis testing - what do we do if we get a large p value?

small p value → reject $H_0$

large p value → ?

Large p says that it would be reasonable to see data like our observed data, if $H_0$ were true. So we don't reject $H_0$. But we don't say that we "accept" $H_0$, either

don't "accept" $H_0$

The null hypothesis could be true, but that doesn't mean it's the ONLY possibility.

seat belt example: if 63% of the sample wore seatbelts, then it's possible that the true population percent is 60%

seatbelt ex: $H_0: p = 0.6$ what if $\hat{p} = 0.63$?

$H_0$ could be true

We wouldn't reject the possibility that $p = 60\%$.

But that doesn't mean we think it's EXACTLY 60%.
It could also plausibly be true that p=61% or 62%, etc.

p = 0.61, p = 0.62, etc also possible

Based on the data, we just think it could be true that p = 0.6, not that p HAS to be exactly 0.6.

large p → $H_0$ could be true

We often phrase that conclusion by saying that we "fail to reject $H_0$".